

SANDEEP NITHARWAL

Junior Undergraduate

Department of Computer Science and Engineering

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - 2025	B.Tech	Indian Institute of Technology Kanpur	8.53/10
2021	CBSE (XII)	Jaipuria Vidyalaya, Jaipur	94.6%
2019	CBSE (X)	Jaipuria Vidyalaya, Jaipur	92.8%

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 1** in **India Terminal** by **Correlation One** and **Citadel|Citadel Securities** (2023)
- Attained position among top **20** in **AlphaVerse Premier League** by **AlphaGrep Securities** (2023)
- Secured **All India Rank 810** in **JEE Advanced** and **All India Rank 1904** in **JEE Mains** (2021)
- Received **KVPY Fellowship** awarded by **IISc Bangalore** by securing **All India Rank 1399** (2021)
- Secured place in **Top 90** in **IOQM**, qualifying for **Indian National Mathematical Olympiad (INMO)**(2021)
- Cleared the **MAS** cutoff of **Indian Olympiad Qualifier (IOQ)** Physics, Chemistry and Astronomy (2021)
- Secured place in **Top 300** in **PRMO**, qualifying for **Regional Mathematical Olympiad (RMO)** (2019)
- Secured **All India Rank 22A** in **National Mathematics Talent Contests (NMTC)** (2018)

WORK EXPERIENCE

Internship at Vizudara | Mathematics Content Developer | Dr. Raj Dandekar, Co-founder, Vizudara (Nov'22-Dec'22)

- Created **20+** animated concept videos on **Circles**, **Constructions** and **Triangles** for in-depth **visual learning**
- Utilized **Manim** library of **Python** to develop accurate animations effectively conveying mathematical concepts

KEY PROJECTS

Unified Portal for Hall Automation | Course Project | CS253 | Prof. Indranil Saha |  (Jan'23-May'23)

- Designed and implemented software solution to automate various Hall services at IIT Kanpur's Halls of Residence
- Followed **Model View Controller (MVC)** architecture pattern and the **Waterfall** model during development
- Designed the **frontend** using **Figma** and implemented the **backend** functionality using the **Django** Framework
- Conducted thorough **unit testing** with the **Django Test** Framework and **integration testing** with **Selenium**

Introduction to Machine Learning | Course Project | CS771 | Prof. Purushottam Kar |  (Jan'23-May'23)

- Mathematically analyzed and proved the vulnerability of Advanced XORRO PUF to single linear model decryption
- Achieved **99.2%** accuracy by implementing feature-rich **LinearSVC** model on 1040 derived features from original 72 features to crack an **Advanced XOR Ring Oscillator (XORRO) Physical Unclonable Function (PUF)**
- Developed and implemented a highly accurate **decision tree** learning algorithm for word guessing, achieving **100%** accuracy rate with training time of **0.23** seconds, memory size of **1.5 MB** and an average of **3.7** queries

Cervical Cancer Screening Prediction | Course Project | CS253 | Prof. Indranil Saha |  (Mar'23-Apr'23)

- Utilized **SVM** and **KNN** as classifiers, **PCA** for feature extraction, **SMOTE** and **ADASYN** to balance classes
- Visualized data, removed outliers, normalized data and identified correlated features using **correlation heatmap**

CSE Bubble | Course Project | CS220 | Prof. Urbi Chatterjee |  (Jan'23-May'23)

- Designed and implemented efficient processor with **Instruction Set Architecture (ISA)** comparable to **MIPS**
- Built processor having architecture capable of performing single-cycle fetch, decode, and execution of instructions
- Implemented Bubble Sort algorithm in **MIPS assembly** language and executed the machine code on processor

COMPETITIVE PROGRAMMING

- Highest rating **1634** (Expert) on Codeforces (**Sandeep_Nitharwal**), highest rank 317 in Div3 and 495 in Div2
- Placed among the top teams of IIT Kanpur at the ICPC Kanpur-Mathura 2022 Qualifier Round (**Team SAS**)

TECHNICAL SKILLS

- Programming Languages:** C, C++, Python, MySql
- Exposure:** Verilog HDL, MIPS Assembly Language, Javascript, Bash, HTML, CSS, Seaborn, Imblearn
- Utilities:** Git, GitHub, $\text{\LaTeX}$, Figma, Django, Numpy, Pandas, Matplotlib, Manim, QtSpim, Pickle, Scikit-learn

RELEVANT COURSES

Probability for Computer Science	Software Development and Operation	Data Structures and Algorithms
Introduction to Machine Learning	Discrete Mathematics	Computer Organization
Logic for Computer Science	Fundamentals of Computing	Introduction to Electronics
Real Analysis	Ordinary Differential Equations	Linear Algebra